



Glycoconjugates, hypothetical proteins, and post-translational modification: Importance in host–pathogen interaction and antitubercular intervention development

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Abstract

With the emergence of multidrug-resistant bacteria, insufficiency of the established chemotherapy, and the existing vaccine BCG, tuberculosis (TB) subsists as the chief cause of death in different parts of the world. Thus, identification of novel target proteins is urgently required to develop more effective TB interventions. However, the novel vaccine and drug target knowledge based on the essentiality of the pathogen cell envelope components such as glycoconjugates, glycans, and the peptidoglycan layer of the lipid-rich capsule are limited. Furthermore, most of the genes encoding proteins are characterized as hypothetical and functionally unknown. Correspondingly, some researchers have shown that the lipid and sugar components of the envelope glycoconjugates are largely in charge of TB pathogenesis and encounter many drugs and vaccines. Therefore, in this review we provide an insight into a comprehensive study concerning the importance of cell envelope glycoconjugates and hypothetical proteins, the impact of post-translational modification, and the bioinformatics-based implications for better antitubercular intervention development.

KEYWORDS

antigenicity, bioinformatics, cell envelope, glycoconjugates, hypothetical protein, *Mycobacterium tuberculosis*

Key Points

- The *Mtb* cell envelope glycoproteins are the origin of host–pathogen interactions and the main constituents of triggering pathogen drug resistance.
- Approaches of recognizing hypothetical proteins may aid in the credentials of novel proteins for antitubercular drug and vaccine development.
- The PTMs of mycobacterial proteins are acknowledged to modify immunogenicity or pathogenicity and have an impact on the vaccine discovery.

1 | INTRODUCTION

The array of communicable diseases and the risk of rising antimicrobial resistance in an infectious disease like tuberculosis (TB) jeopardize public health throughout the globe. TB is caused by *Mycobacterium tuberculosis* complex

(MTBC) bacteria, principally by the intracellular pathogen *Mycobacterium tuberculosis* (*Mtb*) (Yang et al., 2019). Likewise, bovine TB is responsible for high economic losses in livestock production globally and represents an ongoing threat for zoonotic TB in humans (Brites et al., 2018). The hallmark of *Mtb* is their propensity to cause long-lasting

contagions that resulted in similar pathologies in a range of hosts and a capability to proliferate within host macrophages in the face of its killing effects (Cosma et al., 2003). The MTBC bacteria comprise closely associated pathogens that have more than 99% identity at the nucleotide level. However, they vary considerably in affected host range, syndrome type in trial animals, morphology, and biochemistry (Cosma et al., 2003).

TB was declared a global emergency by the WHO in 1994 but incessantly exists as a public health threat by affecting a large portion of the world population in many low-income countries (Barbier et al., 2018; WHO, 2019). The actively flourishing globalization, migration of people, and rapid urbanization have boosted this airborne disease spread (Medie et al., 2011; WHO, 2019). Several of the existing antitubercular drugs look like to be efficient when directed in the groups. However, in some cases their relationships have been interrogated since the possibility of opposition among medicines (Danelishvili et al., 2017). Consequently, this enforced to the cessation and interruption of treatment further and also leads to the development and spread of drug-resistant TB strains (Liang et al., 2017). On the other hand, the existing TB disease diagnostic techniques cannot differentiate the active TB and dormant TB infection (Coppola & Ottenhoff, 2018). For instance, the ancient tuberculin skin test (TST) and the new interferon-gamma release analyses (IGRAs) of latent TB infection (LTBI) diagnosis methods provide indirect and imperfect information (Auguste et al., 2017). Moreover, the existing licensed TB vaccine Bacillus Calmette–Guerin (BCG) does not adequately inhibit the expansion of pulmonary TB in adults and miss the mark to protect individuals against re-infection (Peng et al., 2012; WHO, 2019). In global prospects, the pathogen anatomical and physiological complexity, along with the chronic nature of *Mtb* infection, is the cause of substantial difficulties in diagnosis, treatment, and controlling TB epidemics (Cook et al., 2009; Danelishvili et al., 2017). Likewise, the alarming growth of various forms of drug-resistant TB such as multiple drug-resistant TB (MDR-TB), extremely drug-resistant TB (XDR-TB), and total drug-resistant TB (TDR-TB) along with the time required for the golden standard diagnostic test is an extra challenge for the management of TB (Liang et al., 2017; Sandgren et al., 2009; WHO, 2019).

Thus, the well-thought-of understandings of the *Mtb* epidemiology, the tubercle bacilli anatomy, and the virulence factors have been suggested as future noble target investigations (Delogu et al., 2013; Van Wyk et al., 2017). Therefore, the key opinion of this review is to present a comprehensive overview on TB pathogen cell envelope glycoconjugates and hypothetical protein (HP) significance on TB disease interventions. Furthermore, the review aims to indicate the significance of conceptual approaches on the “unbiased” genome-wide identification of the noble potent antigen and

discuss on protein post-translational modification influence during infection and importance of its recognition during TB subunit vaccine and drug discovery.

2 | *Mtb* CELL ENVELOPE

The cell wall is a defining characteristic of all bacteria. Maintaining the cell shape, mechanical resistance of the cells and withstanding hostile environments are among its many key functions (Jankute et al., 2015; Maitra et al., 2019). The unique architecture of *Mtb* cell envelope is comprised of high content of a variety of lipids, carbohydrates, glycoproteins, lipoproteins, proteins of unknown function, and other virulence proteins (Catalão et al., 2019; Forrellad et al., 2013). It is vital for cell growth and survival, upholding cellular integrity, and virulence (Abrahams & Besra, 2018; Catalão et al., 2019; Jankute et al., 2015; Maitra et al., 2019). The mycolyl–arabinogalactan–peptidoglycan (mAGP) complex and the phosphatidyl-*myo*-inositol-based lipoglycans are the key features of large macro-molecular complex of mycobacterial cell envelope synthesis and assembly. The unusual architectural complexity of cell envelope structures assembly and their various glycoconjugates components discriminates the genus mycobacteria from other prokaryotes’ species (Grzegorzewicz et al., 2016). The lipid portion is one of the main antigenic constituents triggering immune responses via immune cell pattern recognition receptors (PRRs) such as Toll-like receptors (TLRs) and also triggers pathogen drug resistance (Angala et al., 2014; Matsunaga et al., 2004). The bacilli cell envelope complex structures are the origin of host–pathogen associations and liable to aid as a barrier against several hydrophilic antitubercular chemotherapeutics, promoting a resistance for many commonly used antibiotics (Angala et al., 2014; Catalão et al., 2019; Chiaradia et al., 2017; Jankute et al., 2015). This is due to its very low cellular permeability and the ability to accumulate mutations. Moreover, this unique cell envelope structure outcome has been correlated with *Mtb* to duplicate inside macrophages despite the bactericidal abilities of such cells (Cosma et al., 2003; Daffe & Lemassu, 2000; Maitra et al., 2019; Sebastian et al., 2020). Besides, its very low cellular permeability, possession of a large number of efflux pumps, and its ability to accumulate mutations mark it as the main constituents for triggering pathogen drug resistance effect and insusceptibility to the general chemicals like alkalis and hypochlorite (Angala et al., 2014; Daffe & Lemassu, 2000; Maitra et al., 2019).

In general, the unusual thick tubercle bacilli cell envelope comprises the three major cross-linked structures. The innermost layer is the plasma membrane that appears typical of bacterial membranes except for the presence of mycobacterium-specific glycolipids, lipoglycans, and

lipoproteins. The intermediate, next to the plasma membrane, is the cell wall core. It is comprised of a cross-linked peptidoglycan (PG), a very highly branched arabinogalactan (AG) heteropolysaccharide, and the characteristic long-chain mycolic acid (MA) complex. The external surface, that is embedded with waxy element known as capsular layer containing mostly polysaccharides, glucans, arabinans, and arabinomannans (Angala et al., 2014; Bhat et al., 2017; Jankute et al., 2015; Kaur et al., 2009; Sebastian et al., 2020). The schematic illustration of the *Mtb* cell wall is given in Figure 1. However, the organization and dissemination of the many soluble cell envelope units, such as lipoarabinomannan (LAM), the other phosphatidylinositol-anchored proteins containing glycolipids, the phthiocerol, and trehalose-containing lipids, which are important to the pathogenesis of *Mtb* are poorly understood (Kaur et al., 2009).

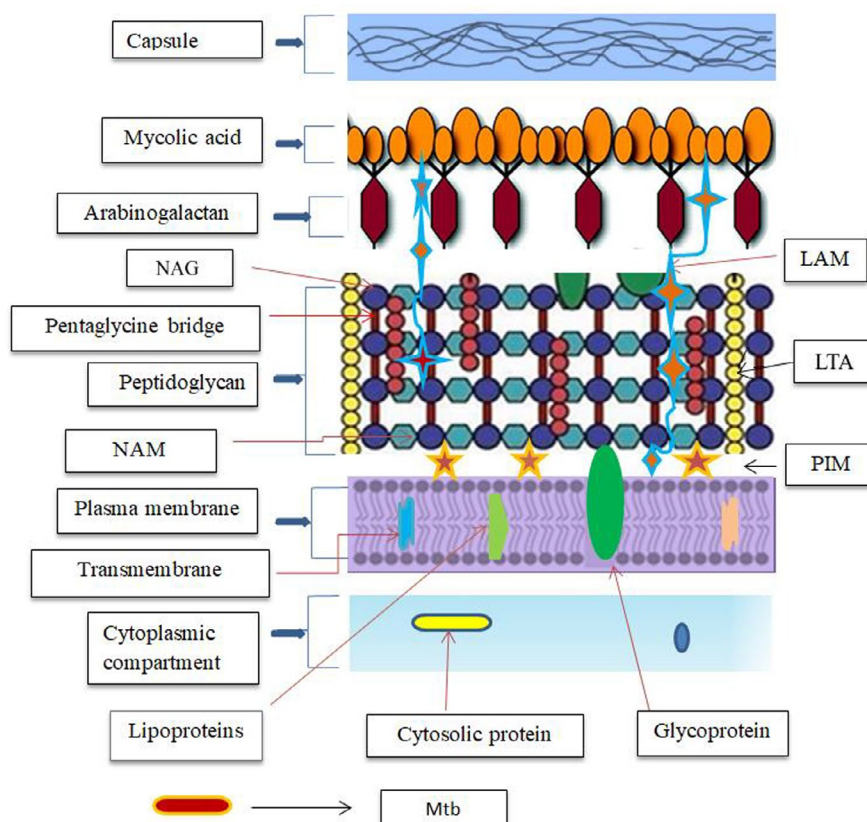
The mycobacterium cell wall arrangement chemical analysis has shown the appearance of wide non-covalently bound glycolipids such as phosphatidyl-myoinositol mannosides (PIMs), lipopolysaccharides (LPSs) such as lipomannan (LM) and lipoarabinomannans (LAM), phthiocerol dimycocerosates (PDIM), trehalose dimycolate (TDM), and mannose-capped lipoarabinomannans (ManLAM) (Daffe, 2008). Additional lipids like sulfolipids (SL, I), polyacyltrehaloses (PAT, II), phthiocerol dimycocerosate (DIM, III), mannosyl-b-1-phosphomycoketides (MPMs, IV), and the mycolic arabinogalactan (MAG) interconnect diacyltrehalose (DAT) are occasionally shed out of the cell wall. The pathogenic *Mtb*

bacilli LAMs are crowned with mannose residues (ManLAM) (Briken et al., 2004). The phosphatidylinositol mannosides (PIMs) are a major class of glycolipids in all mycobacteria and are virulence determinant of *Mtb*. It also involved with diverse roles like host-pathogen interactions, maintaining cell integrity, and the permeability barrier of the cell envelope (Morita et al., 2006). Furthermore, these key components shield the bacteria from its environment in modulating the host response during infection, promote the phagocytosis of *Mtb* through interaction with mannose-specific C-type lectins (C-TLs) receptor, regulate phagosome maturation, and are thus essential for mycobacterial survival (Sonawane et al., 2012). The pathogen cell walls' extremely complex lipid richness indicates its importance to the physiology and virulence of these organisms (Medjahed et al., 2010). This suggests that compounds inhibiting their synthesis could potentiate the existing anti-TB drugs. Regardless of its noticeable significance, little is recognized on the components and organization of mycobacterial cell wall structures (Chiaradia et al., 2017).

2.1 | Peptidoglycan (PG)

The PG is a major structural component of the cell wall of both Gram-positive and Gram-negative bacteria including the majority of the pathogenic bacteria, such as *Mtb* (Abrahams & Besra, 2018; Vollmer et al., 2008). The basic structure of

FIGURE 1 The schematic illustration of the *Mtb* cell wall. The innermost layer next to the cytoplasm is the plasma membrane. The intermediate, next to the plasma membrane, is the cell wall core. It is comprised of a very cross-linked peptidoglycan (PG), a highly branched arabinogalactan (AG) heteropolysaccharide, and the characteristic long-chain mycolic acid (MA) complex. The external surface, which is embedded with waxy element known as capsular layer. Abbreviations: PIM: phosphatidyl-myoinositol mannosides; LAM: lipoarabinomannans; LTA: Lipoteichoic acid; NAG: *N*-acetylglucosamine; and NAM: *N*-acetylmuramic acid



PG is conserved. PG of *Mtb* is made of glycan strands of alternating $\beta(1 \rightarrow 4)$ -linked *N*-acetylglucosamine (GlcNAc) and *N*-acetylmuramic acid (MurNAc) units cross-linked via pentaglycine inter-bridge residues and non-classical (3 $\rightarrow$ 3) transpeptide bonds (Angala et al., 2014; Kaur et al., 2009; Lovering et al., 2012; Nasiri et al., 2017). The mature *Mtb* PG architecture is also marked by a high degree of direct peptide cross-links, which is a characteristic feature that is not frequently found in other bacteria (Catalão et al., 2019).

PG isolated from *Mtb* and *M. smegmatis* contains both *N*-acetylmuramic acid (MurNAc) and *N*-glycolylmuramic acid (MurNGlyc), whereas most other bacteria, including *M. leprae*, solely contain MurNAc (Mahapatra et al., 2008; Petit et al., 1969). The presence of MurNGlyc increases the resistance of mycobacterial PG to lysozyme (Raymond et al., 2005). These arrangements probably support the physical inelasticity which is essential to maintain cellular integrity and to counter the osmotic effects inside the infected host or in different ecological niches. Furthermore, the PG architecture component may allow flexibility in the pathogen cell morphology, which is essential to acclimatize the cell form alterations in the course of several phases of development, segmentation, and cell–cell communication and in the initiation of the host immune response during infection (Maitra et al., 2019).

The PG layer of *Mtb* is surrounded by lipids, carbohydrates, and phosphatidyl-myo-inositol-based lipoglycans that provide a permeability barrier against hydrophilic drugs. The cell wall cores, mycolyl–arabinogalactan–peptidoglycan (MA-AG-PG) (mAGP) complex is the unusual peptidoglycan (PG) layer, that plays a key role in virulence by enabling pathogenic species to survive inside the host and resist antibiotic pressure (Catalão et al., 2019; Maitra et al., 2019). The biosynthesis of PG is dynamic and multifaceted procedure connecting nearly 20 enzyme activities (Lovering et al., 2012). Based on their cellular location, the PG biosynthesis is categorized into three different phases (cytoplasmic, periplasmic, and membrane-associated) (Teo & Roper, 2015). The enzymes intricate in the microbial PG biosynthetic pathway are important as most of them do not have any similarities to the host. This further makes it optimal for selective targeting, subsequently enhancing its potential as a suitable drug target (Mdluli & Spigelman, 2006; Machowski et al., 2014; Kumar et al., 2020). Additionally, their indispensable nature and existence in Gram-negative and Gram-positive pathogens sites make them exceptional desires for broad-spectrum antibiotics. Thus, PG biosynthetic pathway is one of the positive sets of antibacterial targets including tubercle bacilli. This would be possible via targeting the putative enzymes that are involved in catalyzing, recycling, degrading, remodeling, and assembly in the PG biosynthesis pathways which is essential for mycobacteria survival. As a consequence, their inhibition is expected

to result in selective destruction of the bacteria (Abrahams & Besra, 2018; Catalão et al., 2019; Maitra et al., 2019). For instance, the *Mtb* PG biosynthesis phase I pathway crucial enzyme, glutamate racemase (MurI), is used as a target for TB first-line drug ethambutol (EMB) (Bugg & Walsh, 1992; Sengupta et al., 2008; Pawar et al., 2019). The other PG biosynthesis crucial enzymes, alanine racemase (Alr) and D-alanine-D-alanine ligase (Ddl), are used as a target for TB second-line drug, D-cycloserine (Hwang et al., 2013). D-cycloserine is a structural analog of D-alanine that interferes with the PG biosynthesis via inhibition of Alr and Ddl enzyme activity (Catalão et al., 2019; Hwang et al., 2013). The MurG-catalyzed conversion step of lipid I to lipid II PG synthesis pathway blocking is targeted by ramoplanin and enduracidin (Fang et al., 2006). MurG is among the set of Mur family of enzymes, which are involved in catalysis of the cytoplasmic steps of peptidoglycan biosynthesis, specifically the final monomeric block of lipid II producing in PG synthesis (Kumar et al., 2020).

2.2 | sMtb mycolic (Lipoteichoic) acid and capsule importance

The mycolic acids (MAs) are lipids connected to the cell envelope arabinogalactan (AG), which plays vital roles in the conservation of cell wall structures and maintains the virulence of the pathogen (Marrakchi et al., 2014). In addition, the cell envelope hydrophobic properties aid the bacteria to counter many drugs and attributed toward the resistance of the majority of the broad-spectrum antibiotics (Chiaradia et al., 2017). Moreover, the MAs also have a contribution to biofilm creation and thereby used as targets for a fresh anticipative antitubercular nominee hunt (Marrakchi et al., 2014). Interestingly, MA synthesis is the site of action for the most efficient and the frontline TB therapeutic drug, isoniazid (INH) (Takayama et al., 1972).

The capsule of pathogenic mycobacteria such as *Mtb* is mostly poised of polysaccharides and proteins, located outside of the mycolic acid layers and loosely attached with a tiny amount of lipids (Kaur et al., 2009). Bacterial capsules are shielding assemblies on various pathogenic microbes. Also, it has been significant for the attachments of the host and contributes toward the invasion of macrophages during infection (Sonawane et al., 2012). Essentially, the extent and quantities of external cover and capsular resources differ with the *Mtb* categories. These key constituents protect the bacteria from the hostile environment and stimulate the phagocytosis of the bacilli via interconnecting with mannose-specific C-type lectins (C-TLs). Furthermore, it helps to regulate phagosome progress and consequently vital for tubercle bacilli existence (Fenton et al., 2005). This highlights that the knowledge of the constituents of

Mtb membrane has an impact on pathogen physiology and pathogenicity knowledge and further antitubercular intervention development (Chiaradia et al., 2017).

3 | *Mtb* CELL ENVELOPE GLYCOCONJUGATES

The glycoconjugates are anatomically and physiologically essential groupings of the cytosolic and cell envelope-associated macromolecules, such as glycoproteins, LPSs, PG, exopolysaccharides, capsular polysaccharides, lipoglycans, and glycosylated teichoic acids (TAs) (Tytgat & Lebeer, 2014). Cell envelope glycoconjugates are an exciting assembly of molecules that can form a strain-specific pattern on the envelope of the pathogens and are intermediates with the host–pathogen receptor interaction (Catalão et al., 2019). Among the intracellular pathogens, the *Mtb*

infection is mainly transmitted through the respiratory route by inhaling a dozen of tubercle bacilli droplets. Thus, TB primarily affects the respiratory organ lung and produces lung TB (WHO, 2018). The bacilli make a replica within the phagosome of infected alveolar macrophages by blocking the phagosome–lysosome fusion of adapted macrophages (Yang et al., 2019) and escape the killing by macrophages (Cosma et al., 2003; Russell, 2007). The relations and effect of tubercle bacilli in its privileged cell are presented in Figure 2.

Usually, the bacilli possessed macrophages are enclosed inside aggregates of immune cells known as granulomas. On the other hand, in a considerable quantity of conditions approximately 10%, the bacilli leak its restraint and produces vigorous illness (Russell, 2007). In this setting, the bacillus avoids the development of phagolysosome (Deretic et al., 2006) and thereby hinders the macrophage responses to infection such as apoptosis and release of inflammatory cytokines (Miller et al., 2010). These are notable indications

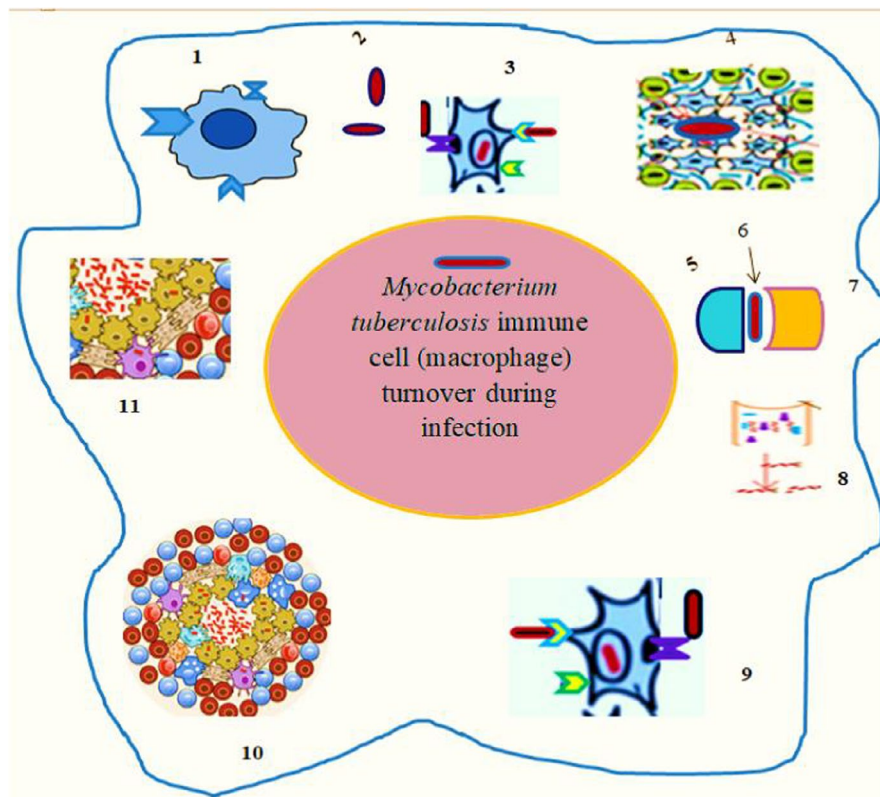


FIGURE 2 The pathogenic *Mycobacterium tuberculosis* bacilli interaction and trafficking in host macrophages. The mycobacteria are inside the phagocytosis immune cell (macrophages) through endocytic pathway and remain within phagosome until the phagosome–lysosome maturation. Subsequently, phagolysosome fusion maturation will proceed to kill the bacilli. Likely, the antigenic peptides are processed and presented to T cells in complex with MHC-II molecules. 1. Immune cell, uninfected macrophage; 2. Free *Mtb*; 3. Infected macrophage; 4. Different immune cells react with *Mtb* (Phagocytic *Mtb*); 5. Phagosome; 6. *Mtb* phagosome & lysosome fusion: at this stage if Immune cell is competent and phagosome & lysosome fusion facilitated *Mtb* will be killed 7). Lysosome; 8). At this phase if some bacilli will persist inside the macrophage and procure to other immune cells; 9). *Mtb* procure new neighbouring Macrophages 10). The infected macrophage at step 8 will procure further immune cells then granuloma will form and the bacilli will come to dormant. 11). If immune system will compromised the dormant bacilli will reactivate, replicate and lung TB will occur. If the case will not treat the bacilli will distribute to other vital organs and miliary disease will occur

of the exceptional glycoconjugate assemblies that exist in the envelope of *Mtb* performing a vital function in the pathogenesis (Ehlers & Daffe, 1998). The selected *Mtb* glycoconjugates in the cell surface are the major sources of bacilli acquaintances, antigenicity, virulence drive, and the immune cell stimulators. Equally, it serves as the main gears for triggering antimicrobial resistance (AMR) in *Mtb*. Consequentially, their upshot drives the chances for emerging “drug resistance TB” which requires new drugs, vaccines, diagnostics, and biomarkers (Angala et al., 2014; Birhanu et al., 2019). On the other hand, the enzymes convoluted in the biosynthesis of mycobacterial envelope glycoconjugates signify possible targets for novel drugs. The glycoconjugates in the innermost layer membranes and in a cytosolic component such as glycogen, glucosylglycerate, polymethylated polysaccharides, and mycothiols are believed to be significant for keeping the cells from osmotic and nitrogen stress, controlling fatty acid synthesis and as a carbohydrate reserve (Behrends et al., 2012; Kaur et al., 2009). In line with their functional role, these multifaceted glycoconjugates produce the main value in modifying the host response in the course of infection (Mishra et al., 2011). They also modify the connection properties of cell coats and extracellular matrixes. This may not be anticipated directly from the expression levels of the enzymes liable for their biosynthesis method.

3.1 | *Mtb* glycoprotein antigenic properties and biological roles

The majority of *Mtb* glycoproteins are involved in cell surface biosynthesis, host–pathogen interaction, fatty acid, and lipid metabolism. Glycoproteins on the layer of the pathogen cell attach to the corresponding host cell receptors and act as key mediators of diverse response. Thus, they have progressed to present their particular sugars and receptors in an approach that obstructs with host glycan-based immune response (Davis & Weiser, 2011). This is signified by microbial pattern recognition, activate the signaling pathways, secretion of bioactive constituents, killing of the bacilli and a decrease or stimulate pathogen colonization possibilities. These strongly interacting pathogen glycoproteins may often a liable target for vaccine and therapeutic drug development. Some reviewers suggest the progress of anti-mycobacterial resistance (AMR) arbitrates to glycoproteins related to drug efflux pumps, drug-hydrolyzing enzymes, or the capacity of modifying *Mtb* cell surface penetrability (Sonawane et al., 2012).

The former *Mtb* researchers reported that glycoproteins impact on host–pathogen interaction and virulence. Similarly, many continual studies forward glycoprotein use in regulating host immune reactions and also identified as

putative virulence factors (Li et al., 2018). For instance, the mannose containing 19-kDa (Rv3763) culture filtrate glycoprotein antigen showed the vital role as the main adhesion protein that interacted with mannose receptors on human T helper 1 (THP-1) cells producing in augmented phagocytosis of *Mycobacteria* (Diaz-Silvestre et al., 2005). Furthermore, glycoproteins involved in PG biosynthesis, such as Mur enzymes and penicillin-binding proteins (PBPs), are possible drug targets (Lovering et al., 2012). The *MTBC* bacteria proteins such as PhoP254 and Ag85A utilized in the biosynthesis of cell wall lipids have a role in the phagosome arresting (Katti et al., 2008). Additionally, different researchers experimentally investigated and reported a list of the *Mtb* glycoproteins with their important roles as shown in Table 1. Even with such vital significances of these proteins in *Mtb* pathogenesis, our present knowledge of *Mtb* glycoprotein is still incomplete. Besides this, a little secreted and cell surface allied glycoproteins have been defined to date (Birhanu et al., 2019; González-Zamorano et al., 2009). Thus, the credentials of glycoprotein in the pathogenesis are significant for the improvement of novel chemotherapeutics or vaccines to encounter the disease. Furthermore, many of the bacterial glycoproteins have an enzymatic activity. Despite their inherent thermostability and resistance toward proteolytic degradation, many of them have a potential role of industrial importance (Upreti et al., 2003). Interestingly, some pathogen glycoproteins have been shown as immunodominant antigens with a role as virulence factors implicating infectivity. For instance, the glycoprotein molecules of S-layer proteins are potentially suitable for immune diagnosis and vaccination (Espitia & Mancilla, 1989). Similarly, *Mtb* glycoproteins perform an essential role in several biological activities. This includes cell attachments and invasion, protein consistency, position, and upkeep of protein conformation. Their roles like cellular signaling, antimicrobial resistance, immunomodulation, intracellular bacterial existence, biofilm development, protein complex formation, antigenicity, pathogenicity, and virulence are influenced by glycosylation (Birhanu et al., 2019). Thereby, the cell wall glycoconjugates signify interphases that have been reflected and achieved to disclose possible drugs and vaccine target nominees.

3.2 | *Mtb* lipoprotein antigenic properties and biological roles

The lipoproteins (LPs) in all bacteria comprising *Mtb* characteristically have an N-terminal signal peptide and lipids connected to a critical cysteine residue. Its acylation permits them to anchor and/or sort to the cell envelope which leads the protein to plasma or outer membrane (Becker & Sander, 2016; Li et al., 2018). However, only limited

TABLE 1 Glycoproteins recognized from *Mtb* roles in regulating host immune reactions and also the identified putative virulence factors reported by different researchers

Gene name	Protein name	Virulence, pathogenesis, and immunogenicity functions	Reference
Apa modD (Rv1860)	Alanine and proline-rich secreted protein 45/47 kDa glycoprotein	Immune-dominant glycoproteins that have potential as a vaccine against TB. Essential for the antigenicity in T cell-dependent immune responses. Deglycosylation of the protein resulted in a 10-fold decrease in the elicitation of a delayed-type hypersensitivity response in guinea pigs and was 30-fold less active in stimulating T lymphocytes. Mediates attachment to the host cell through interaction with pulmonary surfactant protein A	Lara et al. (2004), Ragas et al. (2007)
hbbA (Rv0475)	Heparin-binding hemagglutinin (Adhesin)	Encouraging the adhesion of the mycobacteria to the epithelial cells and fibroblasts. Important for non-phagocytic cell interaction & the pathogenesis of tuberculosis by extrapulmonary. Immunogenic protein may be playing an important role immune protection against tuberculosis. Persuades mycobacterial aggregation which gives the bacteria the aptitude to form a primary biofilm	Pethe et al. (2001), Menozzi et al. (1996)
Fbp complex FbpB & FbpC	Fibronectin-binding protein. Mycolyltransferase/acyltransferase Fructosebiphosphate aldolase, FBP aldolase	Essential in the virulence pathogenesis of <i>Mtb</i> . Its ability to bind FN promotes the adhesion of mycobacteria to the mucosal surface, thus facilitating its entry into the host cells. Utilized in the biosynthesis of cell wall lipids have a role in the phagosome arresting. Moreover, T- and B-cell antigens could have a use in sero-diagnostics	Steingart et al. (2009), Katti et al. (2008)
PstS-1, PstS-2 & PstS-3	PstS complex protein, ABC phosphate transporters	Involved in phosphate import during starvation, a condition prevailing inside the phagosome plays a role in virulence. Pst-S1 and Pst-S3 expected consideration as vaccine nominees	Rengarajan et al. 2005, D'Souza et al., (2002)
DrrC (Rv2938), MmpLs Rv0194, Rv2994 & Rv1273c	Probable doxorubicin resistance ABC transporter permease protein DrrC, MmpLs (mycobacterial membrane protein large) Trehalose monomycolate exporter (MmpLs)	An efflux pumps for antitubercular therapeutics, aid to drug resistance And also MmpLs are interlinked in the transfer of envelope allied lipids and siderophores, and are good-looking pharmacological targets	Da Silva et al., (2011), Chalut (2016)
BlaC Rv2068c	β -lactams and BlaC hydrolyzes nitrocefin	Hydrolyze nitrocefin and other β -lactams, thereby raising the pathogen drug resistance against different modules of β -lactam antibiotics	Nampoothiri et al. (2008)
<i>Mtb</i> 8.4 Rv1174c	<i>Mtb</i> 8.4	An immune-dominant T-cell antigen and a potential candidate for inclusion in a subunit vaccine against TB	Coler et al. 2001
Rv1818c	PE_PGSR33	Encourages mycobacterial aggregation	Delogu and Brennan (2001)
Rv0285 & Rv0286	The PEE family, Rv0285/PE5 and Rv0286/PPE4	Show an essential role in iron acquisition Tufariello et al. (2016)	Cascioferro et al. (2007), Tufariello et al. (2016)

investigational reports have established proof of acylation (Becker & Sander, 2016; Li et al., 2018; Rezwan et al., 2007; Sutcliffe & Harrington, 2004). In mycobacterium, lipoproteins are positioned in the plasma membrane or external membrane and some are also shed from the membrane. The lipoprotein signal peptidase-A (LspA) is a primary lipoprotein signal peptidase accountable to eliminate the signal peptide

following trans-acetylation and translocation. The peptidase cleavage produces the developed form of the lipoprotein that is anchored in the membrane (Rezwan et al., 2007).

The mycobacterium cell envelope has recognized greater than hundreds of proteins in their cellular assembly including lipoproteins. Depending on the diverse bioinformatics algorithms screened, approximately 2.5% of the genes encoded

by *Mtb* proteome is anticipated to encrypt for lipoproteins (Forrellad et al., 2013; Rezwan et al., 2007; Sutcliffe & Harrington, 2004). Unlike the other bacteria, the vast portion coding ability of *Mtb* is enthusiastic to the producers of enzymes intricate in lipogenesis and lipolysis. Initial in the 1990s, Young and Garbe had recognized four lipoprotein antigens in *Mtb* via detergent phase separation and metabolic labeling (Young & Garbe, 1991). Subsequently, others LPs have been recognized via Triton X114 stage separation joined to mass spectrometry or foretold in silico (Målen et al., 2010; Rezwan et al., 2007). It is also established and approved by directing via mutagenesis investigation of these genes which used to disclose their significance in host–pathogen networking (Forrellad et al., 2013).

Lipoproteins are anticipated to signify an important class of cell surface proteins used in nutrient uptake, signaling and protein degradation, drug transfer, and cell envelope homeostasis. The immune cell–microbe relationship via host cell attachment cap also involved in the virulence and immune-regulatory processes of the major pathogen lipoproteins (Becker & Sander, 2016; Buddelmeijer, 2015; Sutcliffe & Harrington, 2004). Moreover, lipoproteins evidently meet pathogen-associated molecular patterns (PAMP) that are acknowledged by the inborn immune recognition mechanisms (Daffe & Lemassu, 2000; Ray et al., 2013). In addition to their organizational role, LPs might involve as stimulating and preventing ligands of TLRs permitting modulation of the immune reaction to the TB bacilli. Some of the different findings of novel potential LPs in cell envelope-associated glycoconjugates are shown with the different direct and indirect roles. For instance, the *Mtb* probable monoacyl phosphatidylinositol tetramannoside-binding lipoprotein (LpqW) regulates the biosynthesis of mycobacterial cell wall vital lipoglycans (PIM and LM/LAM) (Rainczuk et al., 2012). An additional list of *Mtb* lipoproteins with experimental investigation wide-ranging roles is presented in Table 2.

4 | HYPOTHETICAL PROTEINS AND BIOINFORMATICS

Hypothetical proteins (HPs) are unidentified functioned proteins encrypted by expected open reading frames (ORF), but have no investigational proof of functions (Imam et al., 2019). Despite the bulk quantities of biological records such as genomic sequences, and functional genomic data from high-throughput scrutinize, there is a scarcity in the convenient annotation. Approximately, 40% of proteins in the majority of the bacterial genomes are characterized as hypothetical or functionally unknown proteins. Additionally, only ~1% of the proteins in the existing protein library have the status of investigation-based

annotations (Das & Orenge, 2016). Moreover, the biological importance of proteins encrypted by these genes is unknown. However, in many species, many of them have been expected to contribute to important arrays in the survival of pathogens, microbe pathogenicity, and the progression of associated infectious diseases (Mazandu & Mulder, 2012; Yang et al., 2019). Thus, characterization of theoretical and functionally unknown proteins is important to acquire an improved knowledge of the organism with respect to their functions, since all proteins are a translated gene product that links up with the cellular milieu. In several approaches, they help cell development and function and then achieve a variety of astounding biological purposes in the microbes. These starring roles include structural proteins, enzymes, and the moving of elements inside and between cells (Browne et al., 2009). Likewise, this empowers their sorting into other paths and cascades that help as markers and pharmacological targets during drug discovery, and diagnosis especially in pathogens related to human diseases (Imam et al., 2019; Naveed et al., 2018). For instance, Li et al., (2010) identified 3 noble proteins (Rv1987, Rv3807, and Rv3887) via proteome scale investigation and recommends serving as an essential reference for developing new TB diagnostic methods and more effective tuberculosis vaccines. Furthermore, Sha et al., (2017) reconfirmed the immunogenic potency of the reported noble protein, Rv1987 suggesting as an antigenic candidate for diagnostic methods and TB vaccines.

The pathogenic *Mtb* H37Rv strain was primarily identified in 1905 and is a well-studied strain in tuberculosis research. The *Mtb* H37Rv whole-genome sequence and annotation published and available in 1998 were comprised over 4 million base pairs and approximately 4,000 genes (Cole et al., 1998). The genome of *Mtb* H37Rv was re-annotated at different times by different TB researchers and reported as about quarters have been considered to be functionally unknown, hypothetical proteins (HPs) (Camus et al., 2002; Lew et al., 2011). Likewise, different researchers have been studied the *Mtb* genome to investigate the functional roles of HPs in different aspects of analysis (Doerks et al., 2012; Mazandu & Mulder, 2012; Van Wyk et al., 2017; Yang et al., 2019). However, numerous of *Mtb* proteins were annotated as functionally unknown and hypothetical proteins, and their functional value was yet to be concealed (Kumar et al., 2017). The comprehensive sequence and description of the *Mtb* genome have endorsed several new genomic studies on the physiology and pathogenicity of this pathogen (Smith, 2003). For example, approximately 87% of the *Mtb* open reading frame (ORF) genes were cloned into Isopropyl β -D-1-thiogalactopyranoside (IPTG) inducible expression vectors effectively (Li et al., 2010).

TABLE 2 Lipoproteins recognized from *Mtb* roles in regulating host immune reactions and also the identified putative virulence factors reported by different researchers

LPP & RV Name	Function	Reference
MPT83(Rv2873)	A significant antigen with distinct T-cell responses in humans and mice. In the future of DNA and protein subunit vaccine candidates	Kao et al. (2012)
LprG (Rv1411)	Regulates cell envelope porosity, efflux of drugs and obstructs the phagosome–lysosome networking. Influence in <i>Mtb</i> vulnerability to first-line antitubercular antibiotics.	Becker and Sander (2016)
	Recognized as a ligand for the dendritic cell-specific intercellular adhesion. Serves as a carrier to facilitate assembly or trafficking of glycolipids to the mycobacterial cell wall, in this way contributing to virulence	Carroll et al. (2010)
LpqH (Rv3763)	Modification of its essential assembly by deacylation excludes its immunogenic properties. Disturb of the class II trans-activator appearance causes possible strategies for immune evading.	Gehring et al. (2003)
	Triggers apoptosis, accepting leak and distribution of the tubercle.	López et al. (2003)
	Trigger autophagy in human monocytes and, thus, could exert antitubercular action via autophagy introduction.	Li et al. (2018)
	Act as an adhesin, binding to the mannose receptor of THP–1 monocytic cells and thereby stimulating phagocytosis of mycobacteria and facilitate to induce T cell-mediated immunity. TLR2 agonist mechanism of blocking IFN- γ then down regulates antigen performance to T cells or hinders MHC-II antigen processing and performance in macrophages.	Diaz-Silvestre et al. (2005), Forrellad et al. (2013)
LprI(Rv1541c)	Lysozyme attaching lipoprotein that organizes defense to its host from hydrolyzing activity of lysozyme, so as to raise the resistance of <i>Mtb</i> to macrophage secreted lysozyme.	Li et al. (2018)
LpqT(Rv1016)	Exhibited the capability to stimulate the intracellular existence of <i>Mtb</i> . The experimental contagion of macrophages with refined LpqT protein induces macrophage apoptosis and prevents IFN- γ enthused class II MHC expression on macrophage	Li et al. (2018)
PstS1	Indicating a role in virulence. Ligand with the TLR2 and TLR4. Provoked the activation of ERK1/2 and p38 MAPK and also liable for the initiation of the pro-inflammatory cytokines, TNF- α and IL-6 in human macrophage this show value at initial immune reactions in the course of mycobacterial contagion. Supporting the <i>Mtb</i> intracellular existence. Have a potential value in phosphate utilization in medium with diminutive phosphate standard	Young and Garbe (1991), Jung et al. (2006), Espiti et al., (1992)
LpqY (Rv1235)	Putative sugar ABC transport lipoprotein used in the reutilizing of extracellular trehalose, produced from trehalose-consisting elements generated by the pathogen. It could be important for the growth of the bacilli in macrophage during infection	Rengarajan et al. (2005)
lprN	Exclusively essential for in vivo existence	Sasseti and Rubin (2003)
LppX	Facilitates the conveyance of complicated lipids, the phthiocerol dimycocerosates, to the external sheath where these function as virulence elements. Capable to mask the TLR triggering molecules to impede the TLR signaling	Sulzenbacher et al. (2006), Li et al. (2018)
SodC	Support survival of bacilli in an aggressive situation in the course of latency by counterbalancing toxic ingredients as recommended for superoxide dismutase SodC	Becker and Sander (2016)
Rpf (Rv1009)	Resuscitation-promoting factor proteins stimulate dormant cells to divide. Deletion of results in delayed reactivation from chronic tuberculosis	Tufariello et al. (2006)

(Continues)

TABLE 2 (Continued)

LPP & RV Name	Function	Reference
LppP(Rv2330c)	Required for optimal growth in vivo.	Rengarajan et al. (2005), Sassetti and Rubin (2003)
LprO (Rv2290)		
LprK(Rv0173) LpqT(Rv1016c)		
LpqY (Rv1235)		
LpqZ (Rv1244)		
LprG(Rv1411) LppX(Rv2945c)		

4.1 | The experimentally validated hypothetical proteins and their role

In numerous pathogens, HPs can execute a significant role in the existence of pathogens and the distribution of related communicable diseases (Yang et al., 2019). The HP existence almost in every pathogenic organism and other microorganisms has generated noticeable attention in biomedical uses. These proteins are participating in several metabolic and disease pathways of pathogens associated with both humans and animals. Thus, some of the *Mtb* HP values have been categorized through different laboratory investigation trials. For instance, Rv0079 was established to be a DosR regulon function in a decreased role in protein synthesis and networking with TLR-2 to ratify cytokine secretion (Yang et al., 2019). Sassetti and Rubin (2003) investigated and exposed the two HPs (Rv1099c and Rv0573c) by utilizing the TraSH method that is needed for *Mtb* existence in the course of infection in a mouse model of tuberculosis. The Rv1290c, Rv1891, and Rv3404c proteins are HPs, which have an important use in determining the degree of the virulence of *Mtb* (McAdam et al., 2002). The HP encrypted by the Rv1503c operon has a role in glycolipid metabolism (Brodin et al., 2010). Similarly, PPE68 (Rv3873) might provide vital support in the *Mtb* existence in diverse situations (Chen et al., 2017; Yang et al., 2019). Likewise, Iantomasi et al., (2012) reported the HPs of the PE-PGRS family PE-PGRS30 to have a role in the inhibition of phagolysosomal fusion.

4.2 | Bioinformatics tool benefits in hypothetical protein characterization

In silico annotation of hypothetical proteins is a crucial approach that may aid in the credentials of novel proteins. Moreover, it used for preparation and directing investigation on unknown proteins. Accordingly, a re-annotation effort was implemented by Bhardwaj et al., (2011) with highlights of the glycogenome feature of the *Mtb* genome. This genome re-annotation exposed more than 200 glycoproteins, which have not been formerly described.

It was along with the protein families of ESX, PE-PPE, and PE-PGRS proteins. For instance, ESX, PE/PPE, PE-PGRS, PstS complex, and antigen 85 complex comprise more than 15 lipoproteins and more than 100 hypothetical antigenic proteins. Among these, more than ten of the lipoproteins were similar to *Mtb* H37Rv culture filtrate as reported by Malen et al., (2007) and Gonzalez-Zamorano et al., (2009). Similarly, our in silico prediction was also able to predict several immunogenic proteins as presented in Table 3 (unpublished data). These predicted proteins require necessary rigorous experimentation to check their actual glycoproteins and the status of glycosylated consideration.

In general, these earlier outcomes showed that uncharacterized proteins might also provide significant value in organisms. However, the purposes of many uncharacterized proteins in *Mtb* are still indistinct (Yang et al., 2019). Mostly, the HPs, like the proline–glutamic acid (PE), proline–proline–glutamic acid (PPE) family, and cell surface antigens associated with virulence, are uncharacterized (Mazandu & Mulder, 2012). They have been expected to show an antigenic dissimilarity, and their exact function is unknown (Abdallah et al., 2009). Thereby, the basis for antigenic biodiversity and regulator of the host cell protective mechanisms leftovers is unidentified (Sampson, 2011). It might also disturb the immune reactions by obstructing antigen processing. It is also anticipated to have a vital role in the intracellular existence, and may support its survival in diverse circumstances (Mazandu & Mulder, 2012). It will be significant in the upcoming to appraise the inferences of these antigenic duplicates and variants for mycobacterial pathogenesis. Most of these proteins have no resemblance with others, and are thus solitary to some pathogens (Mazandu & Mulder, 2012). In several cases, they are inimitable and perform a particular function in the microbes. This ratifies that these huge figures of HPs in microbes may give a vital function in their survival and sustainability or may support their fitness in diverse situations. Nowadays, with the advancement of computational science, bioinformatics-based study of the HPs has been proved. Also, it is a helpful way in a better understanding of microbe's metabolic pathways for different new strategies

TABLE 3 Some HPs on our web free bioinformatics tool established in *silico* prediction

Rv name	Protein description	Subcellular local	SigP/SecP	Antigenicity	Remark
Rv0063	Possible oxidoreductase	Extracellular	Both	0.577	
Rv2878c	Soluble secreted antigen MPT53	Extracellular	Both	0.642	
Rv2376c	Low molecular weight antigen MTB12	Extracellular	Both	0.534	
Rv2198c	Probable transport accessory protein MmpS3	Extracellular	SecP	0.776	
Rv2253	Uncharacterized protein	Extracellular	Both	0.575	
Rv38756	kDa early secretory antigenic target (ESAT-6)	Extracellular	Sec	0.569	
mte28	Proline-rich 28 kDa antigen	Extracellular	Both	0.517	
Apa	Alanine and proline-rich secreted protein Apa45/47 kDa antigen	Extracellular	Both	0.584	
Rv1911c	Putative lipoprotein LppC	Extracellular	secP	0.823	These list of predicted lipoproteins are also identified by Målen et al. (2007), Gonzalez-Zamorano et al., (2009)
Rv2599	Uncharacterized (HP)	Extracellular	Both	0.601	
Rv1566	Possible Inv protein	Extracellular	Both	0.626	
Rv1881c	Probable lipoprotein LppE	Extracellular	Both	0.505	
Rv3763	Lipoprotein LpqH 19 kDa lipoprotein antigen	Extracellular	Both	0.976	
Rv2290	Putative lipoprotein LppO	Extracellular	Both	0.782	
Rv0583c	Probable conserved lipoprotein LpqN	Extracellular	Both	0.592	
Rv0483	L,D-transpeptidase 5, LDT (Ldt (Mt5)) lprQ	Extracellular	SecP	0.642	
Rv1270c	Lipoprotein LprA	Extracellular	Both	0.686	
Rv0418	Probable lipoprotein aminopeptidase LpqL (Leucine aminopeptidase)	Extracellular	Both	0.535	
Rv1539	Lipoprotein signal peptidase (Prolipoprotein signal peptidase) (Signal peptidase II, SPase II) lspA	Extracellular	Both	0.630	
Rv1411c	Lipoarabinomannan carrier protein LprG (27 kDa lipoprotein) (Triacylated glycolipid carrier LprG)	Membrane	Both	0.572	
Rv3390	Probable conserved lipoprotein LpqD	Extracellular	Both	0.458	
Rv2171	Protein LppM (Putative lipoprotein LppM)	Membrane	Both	0.697	
Rv2945c	Putative phthiocerol dimycoserate transporter LppX(Lipoprotein Lppx)	Extracellular	Both	0.609	
Rv1368	Putative diacylated glycolipid transporter LprF (Lipoprotein LprF)	Membrane	Both	0.562	
Rv2080	Putative lipoprotein LppJ	Extracellular	Both	0.553	
Rv1886c	Diacylglycerol acyltransferase/mycolyltransferase Ag85B (30 kDa extracellular protein) (Fibronectin-binding protein B, Fbbs B)	Extracellular	SecP	0.585	
Rv3804c	Diacylglycerolacyltransferase/mycolyltransferase(Acyl-CoA:diacylglycerol acyltransferase) Ag85A- fbpA (Fibronectin-binding protein A, Fbps A)	Extracellular	SecP	0.525	

(Continues)

TABLE 3 (Continued)

Rv name	Protein description	Subcellular local	SigP/SecP	Antigenicity	Remark
Rv0129c	Diacylglycerol acyltransferase/mycolyltransferase (Acyl-CoA:diacylglycerol acyltransferase) Ag85C flpC (Fibronectin-binding protein C, Fbps C)	Extracellular	SecP	0.448	
Rv0934	Phosphate-binding protein PstS 1, PBP 1, PhoS1 (38-kDa glycolipoprotein)	Extracellular	Sec	0.569	
Rv0932c	Phosphate-binding protein PstS 2, PBP 2	Extracellular	both	0.776	
Rv0928,	Phosphate-binding protein PstS 3, PBP 3 (Antigen Ag88) pstS3 phoS2,	Extracellular	Both	0.803	
Rv2576c	Uncharacterized protein	Extracellular	SecP	0.599	
Rv1362c	Uncharacterized protein	Extracellular	SecP	0.762	
Rv2575	Uncharacterized protein	Extracellular	SecP	0.716	
Rv3103c	Hypothetical proline-rich protein	Extracellular	SecP	0.638	
Rv1987	Uncharacterized protein	Extracellular	Both	0.622	
Rv1271c	Uncharacterized protein	Extracellular	Both	0.714	
Rv1268c	Uncharacterized protein	Extracellular	Both	0.578	
Rv2635	Uncharacterized protein	Extracellular	SecP	0.702	
Rv2083	Uncharacterized protein	Extracellular	Both	0.576	
Rv1975	Uncharacterized protein	Extracellular	Both	0.749	
Rv0810c	Uncharacterized protein	Extracellular	SecP	0.874	
Rv0278c	Uncharacterized PE-PGRS family protein PE_PGRS3	Extracellular	Both	1.663	
Rv0747	Uncharacterized PE-PGRS family protein PE_PGRS10	Extracellular	Both	1.692	
Rv1068c	Uncharacterized PE-PGRS family protein PE_PGRS20	Extracellular	SecP	1.873	
Rv1325c	Uncharacterized PE-PGRS family protein PE_PGRS24	Extracellular	SecP	1.761	
Rv1651c	PE-PGRS family protein PE_PGRS30	Extracellular	Both	1.43	
Rv1840c	Uncharacterized PE-PGRS family protein PE_PGRS34	Extracellular	SecP	1.591	
Rv1983	Uncharacterized PE-PGRS family protein PE_PGRS35	Extracellular	SecP	0.897	
Rv2591	Uncharacterized PE-PGRS, family PE_PGRS44	Extracellular	SecP	1.501	
Rv2634c	Uncharacterized PE-PGRS family protein PE_PGRS46	Extracellular	SecP	1.985	
Rv2741	PE-PGRS family protein PE_PGRS47	Extracellular	Both	1.68	
Rv3508	Uncharacterized PE-PGRS family protein PE_PGRS54	Extracellular	SecP	2.6431	
Rv3653	PE-PGRS family protein PE_PGRS61	Extracellular	SecP	1.512	
Rv0151c	PE family protein PE1	Extracellular	-Ve	0.505	
Rv0442c	Uncharacterized PPE family protein PPE10	Extracellular	SecP	0.807	
Rv0755c	Uncharacterized PPE family protein PPE12	Extracellular	SecP	0.821	

(Continues)

TABLE 3 (Continued)

Rv name	Protein description	Subcellular local	SigP/ SecP	Antigenicity	Remark
Rv0878c	Uncharacterized PPE family protein PPE13	Extracellular	SecP	0.749	
Rv1430	Uncharacterized PPE family protein PPE16	Extracellular	-Ve	0.487	
Rv1753c	Uncharacterized PPE family protein PPE24	Extracellular	SecP	0.646	
Rv1548c	Uncharacterized PPE family protein PPE21	Extracellular	SecP	0.697	
Rv1917c	PPE family protein PPE34	Extracellular	SecP	0.685	
Rv2356c	Uncharacterized PPE family protein PPE40	Extracellular	SecP	0.889	
Rv3533c	Uncharacterized PPE family protein PPE62	Extracellular	SecP	0.881	
PPE54	Uncharacterized PPE family protein PPE54	Membrane	SecP	0.6189	
Rv3810	Exported repetitive protein (Cell surface protein PirG) (EXP53)	Extracellular	both	0.673	
Rv2450c	Resuscitation-promoting factor RpfE	Extracellular		0.8	
Rv2389c	Resuscitation-promoting factor RpfD	Extracellular	SecP	0.62	

Note: RV name—*Mycobacterium tuberculosis* genome sequence code, subcellular local—subcellular localization, SigP—SignalP, and SecP—SecretomeP.

in a drug, vaccine, and diagnostic tool development against pathogens and disease control (Bahrami et al., 2019; Brown & Wright, 2016).

5 | *Mtb* POST-TRANSLATIONAL MODIFICATIONS

Post-translational modifications (PTMs) are proprietary to produce a significant function in sustaining protein role, regulation, gesturing, and an excess of extra vital cellular doings in all domains of life (Soufi et al., 2012). The bacteria contain various kinds of PTMs, comprising those usually existing in eukaryotes, such as phosphorylation (Pereira et al., 2011), acetylation (Jones & O'Connor, 2011), glycosylation (Benz & Schmidt, 2002), ubiquitination and glutathionylation (Dalle-Donne et al., 2009). Furthermore, an extra protein alteration of an ubiquitin resembling alteration technique called pupylation is also recognized in *Mycobacteria* species. In pupylation, a slight protein Pup (prokaryotic ubiquitin-like protein) is moved onto the lysine residues of expected proteins and allows them to be recognized and degraded (Iyer et al., 2008). They are recognized to perform a pronounced function in virulence in numerous microbial pathogens. The *Mtb* pathogen interrupts the eukaryotic host macrophage protection by phosphorylating several components within the microbes and disturbing functional processes vital for the subsistence during host infection (Alber, 2009). Besides to show a function in virulence, they help to control several events from protein secretion (Mougous et al., 2007) to cell envelope fusion and dissection (Fiuza et al., 2008). It also aids to control several significant metabolic approaches from carbohydrate to amino acid metabolism (Pereira et al., 2011).

Current progresses in mass spectrometry-based proteomics have emphasized the incidence of many types of PTMs in proteomes of prokaryotes including *Mtb*. The method confidently recognized the presence of serine, threonine, and tyrosine (S/T/Y) phosphorylation, and lysine acetylation on various microbial proteins. The recognized PTMs in *Mtb*, glycosylation, lipidation, or phosphorylation are known controllers of protein physiology or compartmentalization. The degree of phosphorylation within the diverse microbes varies highly. This inconsistency might partially be due to the genome proportions, intricacy, and ecological aspects that the microbes come across. At present, *Mtb* is the bacilli with the greatest recognized S/T/Y phosphorylation activities and consists of 500 phosphorylated locations on almost 300 proteins (Prisic et al., 2010). Additional mycobacteria PTMs, methylation, acetylation, and pupylation are involved and establish protein stability (van Els et al., 2014; Young & Garbe, 1991). Although every PTM increases erraticism to the *Mtb* proteome, comparatively little is known on their significance in the anti-*Mtb* immune reactions (van Els et al., 2014).

5.1 | Post-translational modifications role in *Mtb* intervention discovery

The PTMs of mycobacterial proteins are also recognized to modify immunogenicity or pathogenicity and have high performance on the product of vaccine discovery role. Methodical investigations of Dobos and co-workers earlier approved the existence of true glycosylated proteins in mycobacterium and also recognized several glycoproteins in diverse cellular localization and culture filtrate (CFP) fractions (Dobos et al., 1995). Lara et al., (2004) have also reported that deglycosylation of Rv1860 antigen produced a ten times reduction in the production of a delayed-type hypersensitivity reaction in guinea pigs and was thirty times less active in initiating T lymphocytes. The 45/47 kDa alanine–proline-rich antigen (APA) was the most examined glycoproteins among the mycobacterial glycoproteins and is referred as vaccine nominees (Lara et al., 2004; Ragas et al., 2007). Its immunogenicity was reported to be reduced with alterations in the glycosylation pattern (Romain et al., 1999). Similarly, Temmerman et al., (2004) stated that the methylated HBHA (heparin-binding hemagglutinin) delivered defense score on *Mtb* challenge in mice, while the wild HBHA did not provide protection. Hence, the methylation of the *Mtb* HBHA was recognized to be vital for efficient T-cell immunity (Temmerman et al., 2004). The mutagenesis of anticipated O-glycosylation site in an *Mtb* 19 KDa antigen stimulated proteolytic modification from a cell-associated lipoprotein to a non-acylated secreted antigen (Herrmann et al., 1996). Thus, PTM of proteins showed importance for their aptitude to boost defensive immunity to communicable maladies like TB will become new targets for therapeutic and prophylactic measures.

6 | KEY POINT OF THE REVIEW

This review paper is being distinct from the previous review articles. The former *Mtb* researchers and/or reviews revealed the restricted scientific information on a specific topic. However, our review perceptions were presented based on comprehensive and organized key information on interconnected topics to the scientific community. This enables to accentuate an inclusive timely significant update especially for the TB disease intervention researchers that help to differentiate the failure, trace out the gap to revise, and modify the research scheme. As a result, the key points of this review are the *Mtb* remarkable outer coat glycoconjugates and its novel biochemical mechanisms enable to facilitate its survival under hostile environment. The *Mtb* cell envelope glycoconjugate proteins are the main determinant for pathogenicity, antigenicity, and immunogenicity, and drug resistance effects. Also, it does largely encounter too many drugs

and vaccines targets. Similarly, the various glycoproteins and lipoproteins possible applications in vaccine development, pharmaceuticals, and diagnostics are also touched upon. The review article also points out the influence of PTM on levels of the bacterial physiology. These insights may expand the current understanding of the nature and diversity of the importance of *Mtb* glycoproteins and lipoproteins, the effect of PTM, and the value of HPs. This may ultimately enable to open new possibility of research for identification of potential vaccine and drug targets that are effective in controlling this drug-resistant pathogen.

7 | CONCLUSION

Since the *Mtb* bacilli acknowledged by Dr Robert Koch in 1882 and WHO declared the disease a global emergency in 1994, TB exists as the chief means of death from a solitary infective pathogen in many low-income countries. Currently, with the alarming growth of various forms of drug-resistant TB (MDR, XDR, and TDR), insufficiency of the existing drug and vaccine is challenging for TB control and eradication. The present tuberculosis interventions such as antitubercular therapy as well as TB vaccines are mainly targeted on the pathogen cell envelop protein, secretory proteins, and metabolic and enzyme pathways. Thus, there is an exigent requisite for the progress of an enhanced diagnostic tools, therapeutic remedies, and vaccines. This can be accomplished through the impeding of the tubercle bacilli cell envelope proteins with efficient chemotherapy and consequently by obstructing the contagion practice of the pathogen via potent vaccine. In this review, the most probable potent antigens of the cell envelope glycoconjugates proteins, hypothetical proteins, and the effect of those targeted protein post-translational modifications situations were discussed. The discussion also comprehensively highlighted the role of these proteins on disease progression, circumvention to host immune reaction scheme, and their consequences as TB interference targets. This will be helpful to make the search for antitubercular intervention candidates and the development of chemotherapy or vaccines that demand a logistic challenge.

ETHICS STATEMENT

The authors confirm that the ethical policies of the journal, as noted on the journal's author guideline page, have been adhered to. No ethical approval was required as this is a review article with no original research data.

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CONFLICT OF INTEREST

The authors state that the review was performed in the absence of a recognized potential interest involvement in any entity.

AUTHOR CONTRIBUTIONS

Aregitu Mekuriaw wrote the manuscript and revised the manuscript based on the editor feedback. Rajani Kanta Mahapatra made contributions to scientific discussion on early versions reviewed and provides feedback, and also edits the final manuscript.

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